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Tracking Natural Disasters from Space:

An Exploratory Data Analysis of NASA EONET Events and FIRMS Fire Detections

I. Introduction

1. Project Aims

Human beings are often reminded of how small they are in the face of nature. Natural hazards such as wildfires, volcanic eruptions, floods, and storms have long been a source of destruction, causing severe damage to ecosystems, infrastructure, and human life. Although modern technology has improved our ability to observe the Earth, many natural disasters still occur suddenly and remain difficult to predict with full accuracy. For this reason, it is important to study natural hazard data in a systematic way.

The aim of this project is to **collect and analyze historical data on natural hazard events** in order to better understand where these events occur, how frequently they happen, and what patterns they show over time and across regions. By identifying the characteristics of disaster-prone areas, this project hopes to provide a clearer picture of global hazard activity and support future efforts in disaster preparedness, risk awareness, and environmental monitoring. Rather than viewing natural hazards as isolated events, we approach them as part of larger spatial and temporal patterns that can be explored through data analysis.

2. Data Sources: NASA EONET and NASA FIRMS

This project uses two public data platforms from NASA: the **Earth Observatory Natural Event Tracker (EONET)** and the **Fire Information for Resource Management System (FIRMS)**. Because both are supported within NASA's Earth observation data system, they offer a high level of scientific reliability and public credibility. EONET is an open-source API that provides curated, continuously updated, near real-time metadata on natural events around the world, while FIRMS delivers near real-time active fire and thermal anomaly data based on satellite observations.

EONET focuses on natural hazard events at the event level. It brings together information on categories such as **wildfires, volcanoes, severe storms, floods, droughts,** and other large-scale environmental events, making it useful for studying broad spatial and temporal patterns of disasters. FIRMS, by contrast, is more specialized and detailed: it uses **MODIS** and **VIIRS** satellite observations to detect active fires and thermal anomalies, providing hotspot-level fire records with much finer spatial and temporal detail.

II. Main Data Analysis

1. Get ready the main dataset: EONET

For the main data, we obtain natural hazard event data using the NASA EONET API, which provides continuously updated information on global natural events. The returned data contains a list of events, each with several key attributes. By inspecting the dataset, we get 8 variables from the raw data and split them into 17 features for further analysis, for example:

- **Id**: unique identifier
- **Title**: event name
- **Categories**: types of natural hazards, such as wildfire or volcano

- **geometry:** featured by **coordinates** and **timestamps**
- **magnitude**: intensity of the events featured by **value and unit**
  ...

Among these variables, the **event categories serve as the primary target variable**, as the project focuses on identifying patterns in specific hazard types—particularly wildfires and volcanic eruptions—across geographic locations, time, and intensity indicators such as magnitude.

1.1 Obtain Data from NASA-EONET

The API returns data as a nested dictionary (JSON object) rather than a flat table. Each event contains embedded fields such as categories, sources, and geometry, which include detailed information about the event's type, date, magnitude, and coordinates. Therefore, the raw response needs to be parsed before it can be analyzed with pandas.

1.2 Identify and narrow down the Target

This project aims to study the effect of different variables like location or time on a certain type of natural disaster, which means the **Target** of this study, based on the raw data, is **categorized**. Due to the **limited sample size** for Volcanoes, Sea and Lake Ice, this project mainly focuses on **Wildfires** as the target.

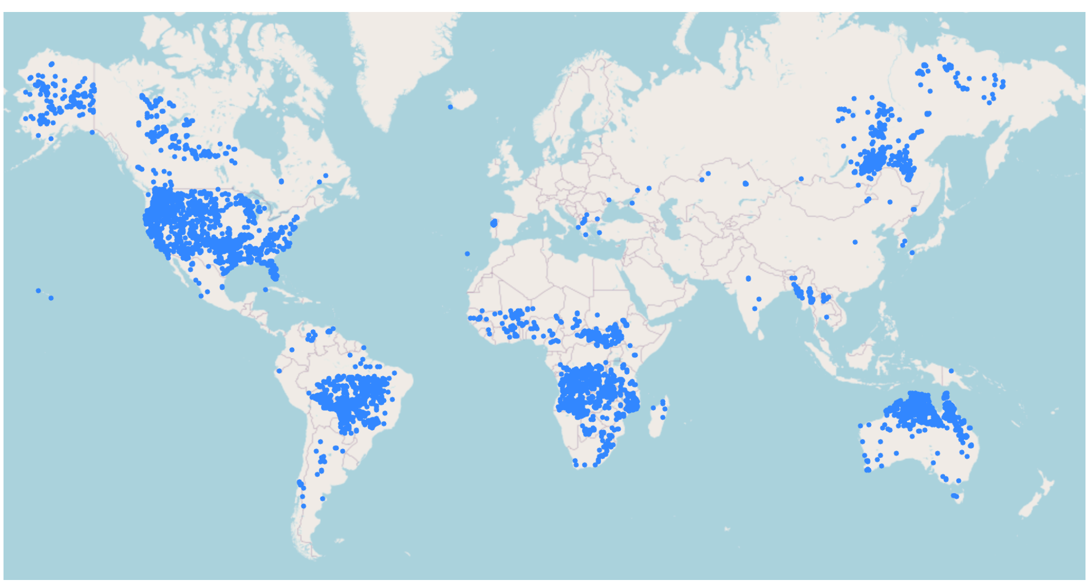
1.3 Build up the Main Data Frame

We further transform the raw nested EONET response into a structured main DataFrame for analysis. The resulting DataFrame contains event-level and geometry-level variables, including event ID, title, category, source count, event date, and geographic location, which provides a clear foundation for later visualization and statistical analysis.

2. Analysis of main data

2.1 Wildfire Spatial Distribution (Visualize Target vs. coordination)

The densest concentrations appear in **North America** (especially the United States and western Canada), **the Amazon basin** in South America, **central and southern Africa**, **eastern Siberia**, and **Australia**.



The clustering also highlights several global wildfire "hotspots," indicating that wildfire risk is **concentrated in specific ecological regions** rather than evenly distributed worldwide.

Overall, the visualization suggests that wildfire activity is strongly associated with **vegetation availability and regional climate patterns**, particularly in tropical forests, savannas, and temperate forest zones.

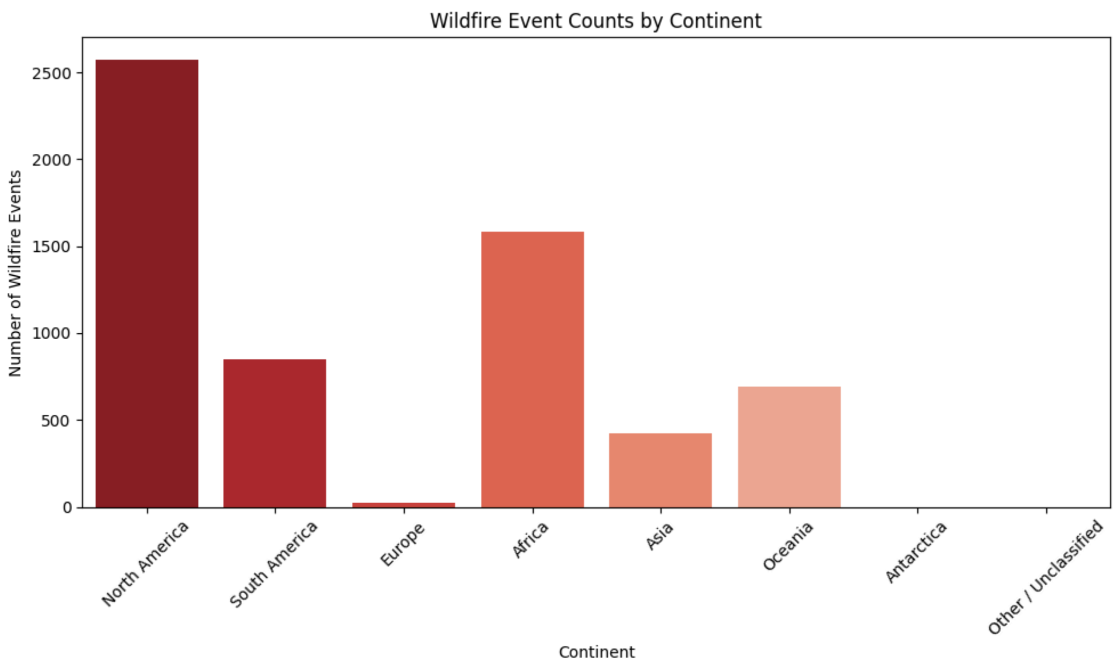
2.2 Wildfire Count by Continent (Target vs. Continent)

The highest wildfire event counts appear in **North America**, followed by **Africa**, with substantial activity also recorded in **South America** and **Oceania**. In contrast, **Asia** shows a lower count, while **Europe** contains comparatively few wildfire events in this dataset.

The distribution highlights several broad continental wildfire "hotspots," suggesting that wildfire activity is **concentrated in specific world regions** rather than evenly distributed across continents. This pattern is consistent with the spatial clustering observed in heavily forested, savanna, and dryland regions.

Overall, the table suggests that wildfire occurrence is strongly shaped by **regional environmental conditions** as well as the **geographic coverage of the NASA dataset.**

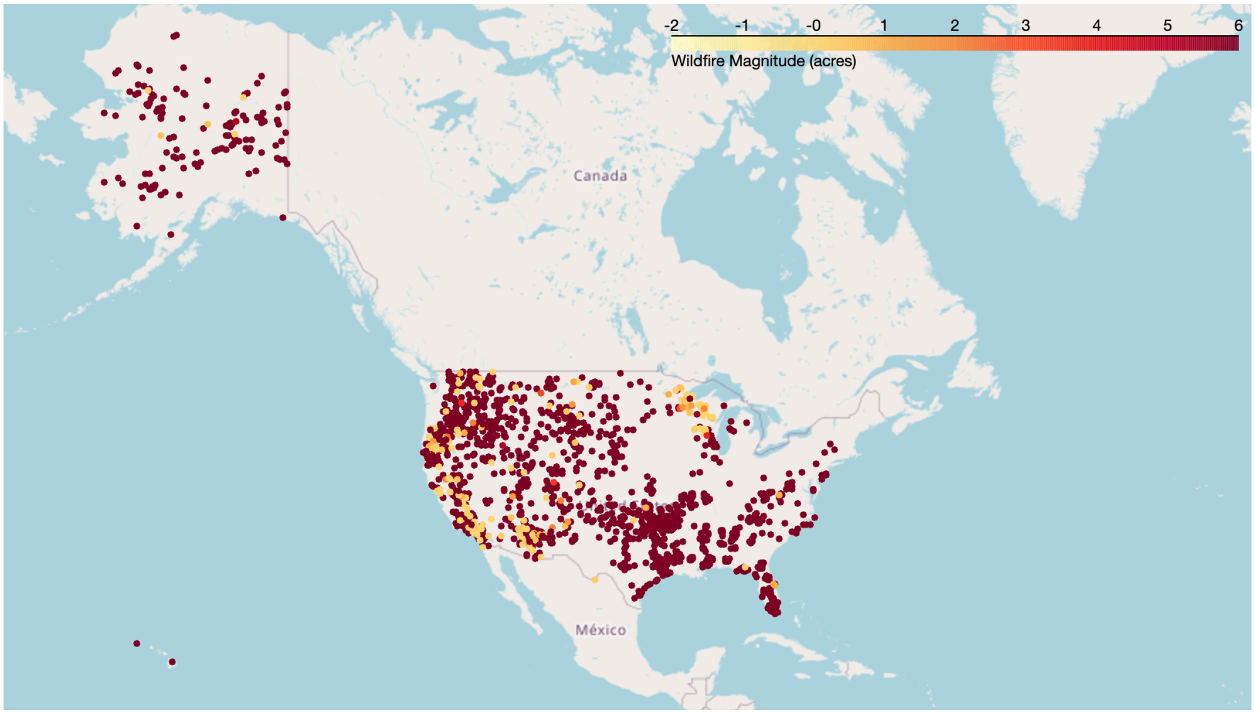
*Notes**:** Since there is no formal continent classification directly defined by latitude and longitude, this study assigns continents using **approximate latitude–longitude** bounding boxes. Records outside these predefined geographic ranges are labeled as **Other / Unclassified.**



2.3 Wildfire Magnitude Distribution (Target vs. Magnitude)

When examining wildfire events with **available magnitude data**, the majority of usable observations are concentrated in the **United States**, which largely reflects the underlying data sources that report wildfire sizes in acres. The map shows that most wildfire events with recorded magnitudes cluster across the **western and southern United States**, with particularly dense activity along the **West Coast and interior regions**.

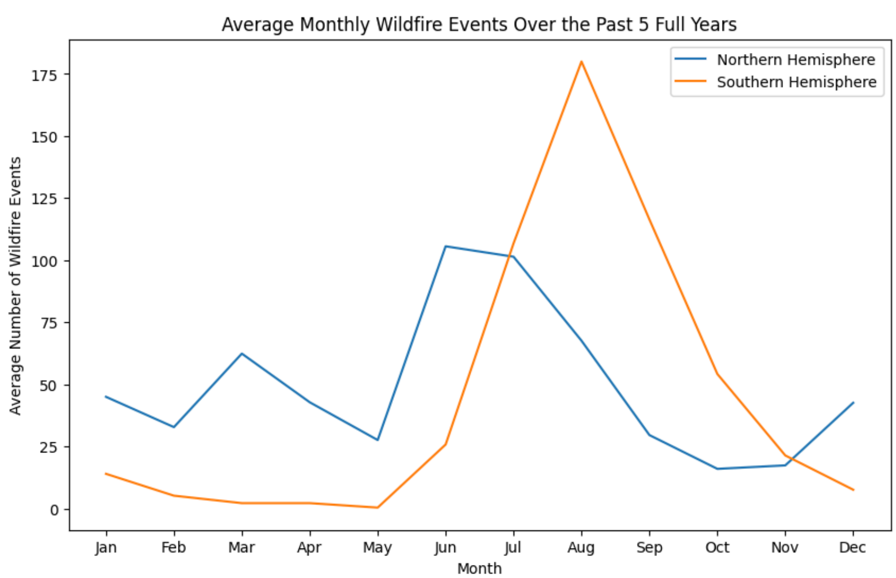
Overall, the visualization highlights the **regional variation** in wildfire size across the United States.



2.4 Monthly Wildfire Seasonality by Hemisphere (Target vs Time & Hemisphere)

Over the past several full years, wildfire activity shows a clear **seasonal pattern** that differs across hemispheres. In the **Northern Hemisphere**, wildfire events tend to **rise during the middle of the year and peak in the warmer months**, while the **Southern Hemisphere** shows relatively higher activity later in the calendar year, reflecting its **opposite seasonal cycle**. This contrast suggests that wildfire occurrence is strongly influenced by climate and seasonal dryness rather than being evenly distributed throughout the year.

Overall, the comparison highlights that global wildfire patterns are **seasonal**, but the timing of peak activity **reverses between the two hemispheres**.



III. Secondary Analysis: Using FIRMS Wildfire to qualify EONET results

The **VIIRS_SNPP_NRT** dataset from FIRMS provides near–real-time satellite detections of active fires collected by the VIIRS sensor aboard the Suomi NPP satellite, including information on fire **location**, detection **day/night time**, and **fire radiative power (FRP)**, which measures fire **intensity**.

1. Do wildfires happen more during the day or at night?

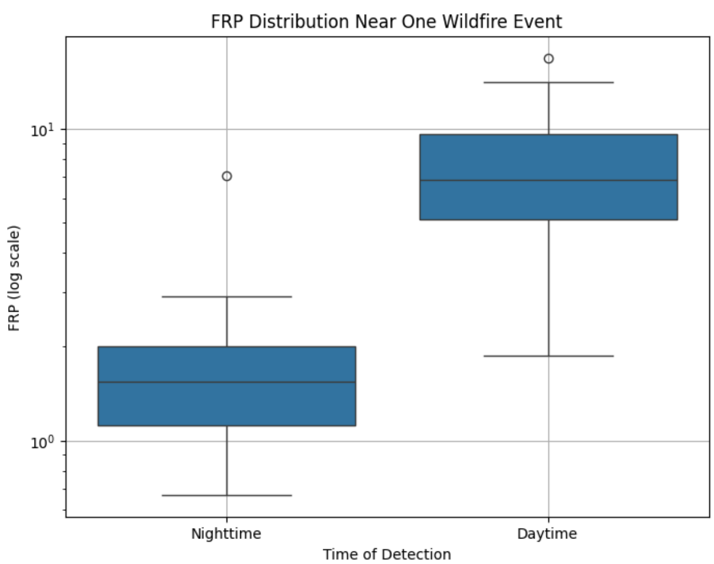
1.1 Finding based on EONET

Wildfire events in the EONET dataset occur **more frequently during the daytime (61%)** than at night (39%), suggesting that recorded wildfire activity is more commonly associated with daytime conditions or daytime satellite detection.

1.2 Finding based on FIRMS

The FIRMS data for the selected wildfire region shows a strong imbalance between daytime and nighttime fire detections, with approximately **87%** of detections occurring **during the day** and only **13% at night**. This suggests that **wildfire activity in this area is much more frequently detected under daytime conditions**. However, this pattern may reflect not only actual fire behavior but also differences in satellite detection capabilities, as daytime observations can capture larger or more visible fires more effectively. Overall, within this localized time window and region, the results indicate a clear dominance of daytime wildfire detections, aligning with the earlier finding based on EONET data.

*Notes: Since the fire spots offered by FIRMS are not necessarily wildfires, the study used the location information from EONET to capture event form FIRMS to align the event type.

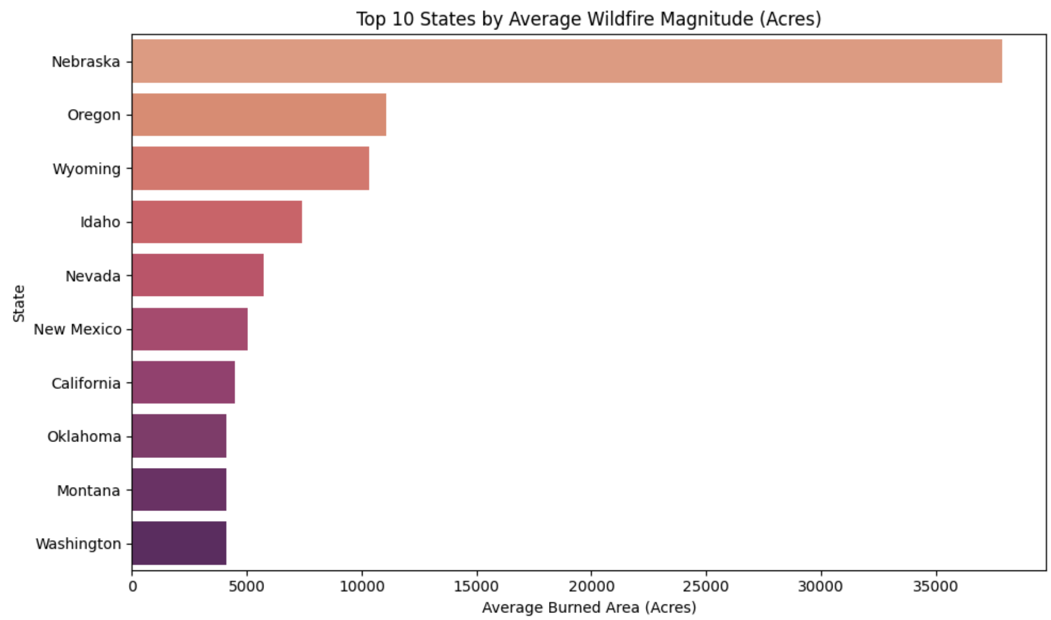


2. How does the Fire Power Radiation vary inside the United States?

The availability of wildfire magnitude data in the EONET dataset is highly **uneven across regions**, with most magnitude measurements concentrated in North America. In response to this data feature, the study zooms in to see the fire intensity variation inside the US by comparing using **nagnitude value** from EONET and **FPR** from FIRMS.

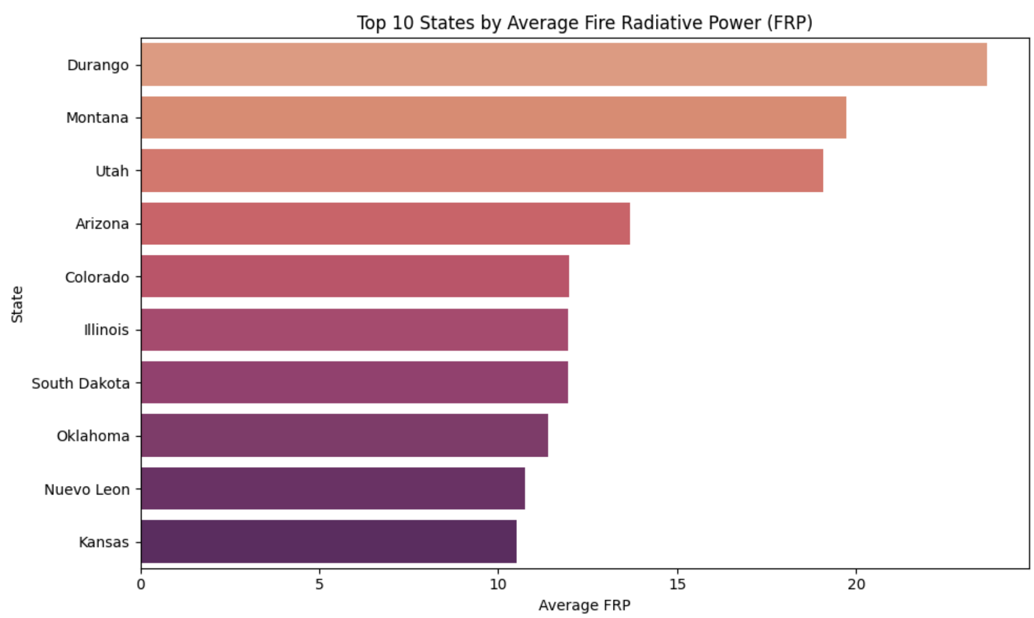
2.1 Finding based on EONET (Wildfire event by Magnitude)

When focusing on U.S. data, clear regional patterns emerge: **western states** consistently exhibit **larger average burned areas**, indicating that wildfires in these regions tend to spread over greater extents. The results show that a small number of states—particularly **Nebraska** and several western states have significantly higher average wildfire magnitudes than others, indicating a **highly skewed distribution** of wildfire size across the United States.



2.2 Finding based on FIRMS (Fire intensity by FPR)

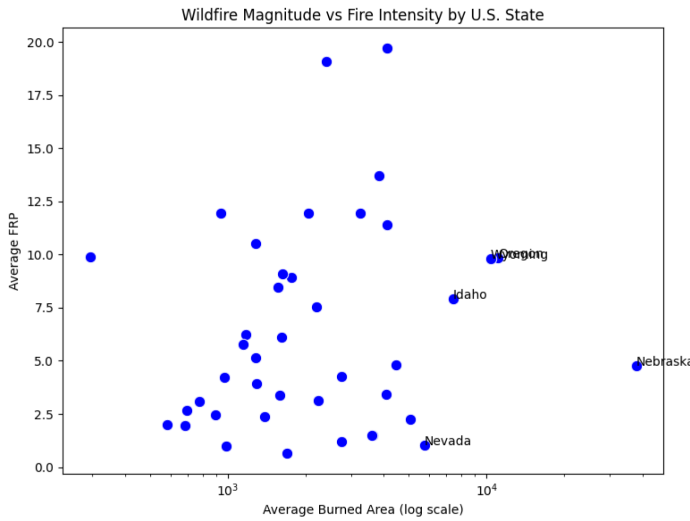
The results indicate that wildfire intensity, as measured by average FRP, varies notably across states, with a few states, such as **Durango, Tennessee, and Utah,** exhibiting substantially **higher fire intensity** than others.



2.3 Associate EONET and FIRMS (Magnitude vs. Intensity)

By combining EONET and FIRMS, a more comprehensive picture of wildfire behavior emerges. While EONET captures the total burned area of wildfire events and FIRMS measures instantaneous fire intensity, their comparison at the state level reveals a generally consistent pattern: **regions with larger average burned areas often also exhibit higher fire intensity.** This relationship is particularly evident in western U.S. states, where both magnitude and FRP are elevated.

However, the relationship is not perfectly linear, reflecting differences in fire behavior, environmental conditions, and data structure between the two sources.

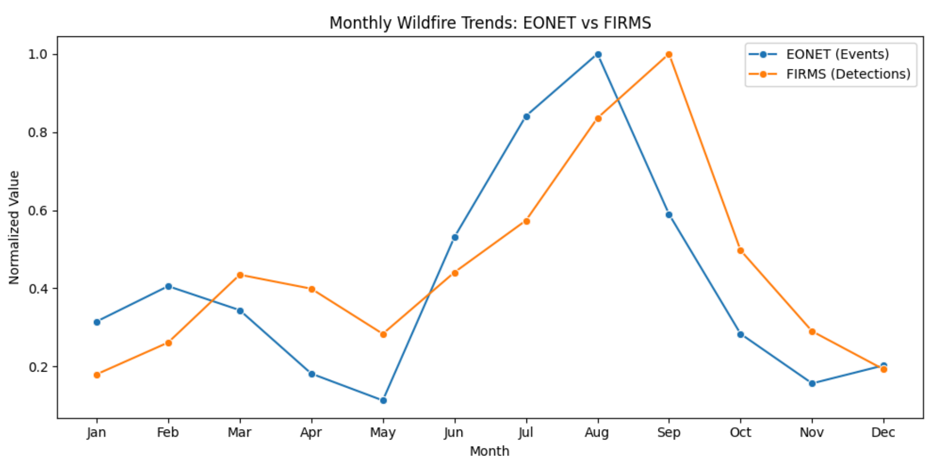


3. What is the seasonal pattern of wildfire?

The results demonstrate that wildfire activity in the United States is **strongly seasonal**, with **late summer and early fall** representing the most active and intense period. The consistency between EONET and FIRMS reinforces the reliability of this pattern, while the addition of FRP highlights that **seasonal factors influence both the occurrence and intensity** of wildfires.

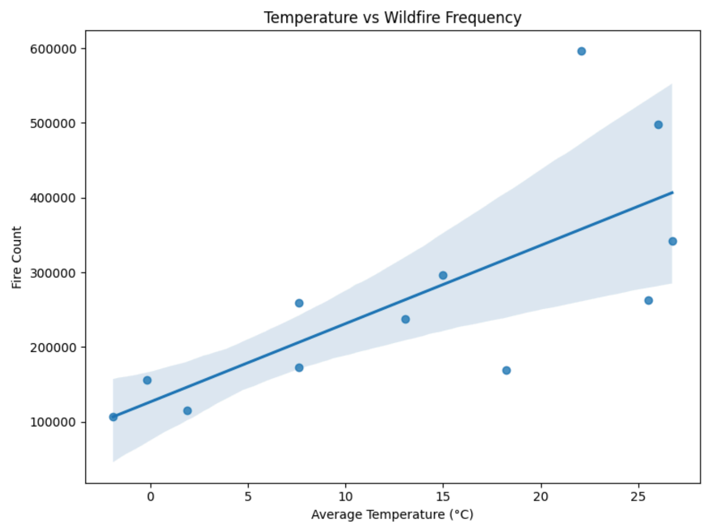
3.1 Combine EONET and FIRMS

Both EONET and FIRMS datasets exhibit **similar seasonal patterns**, with wildfire activity peaking during summer months. This consistency suggests that EONET event reporting aligns well with satellite-based fire detections. Their strong alignment suggests that **both frequency and intensity of wildfires are driven by similar seasonal factors**.



3.2 Seasonal Factor: Temperature

The results show a clear **seasonal temperature pattern**, with temperatures rising from winter to a peak in July–August and then declining toward the end of the year. This pattern closely aligns with wildfire activity, as indicated by the **strong positive correlations between temperature and both fire frequency (0.73) and fire intensity (FRP, 0.81)**. The scatter plot further confirms this relationship, showing that **higher temperatures are associated with substantially higher wildfire counts**. Overall, the findings suggest that temperature is a key driver of both the occurrence and severity of wildfires, reinforcing the idea that seasonal climatic conditions play a crucial role in shaping wildfire dynamics.



IV. Limitations and Possible Improvements

This project has several limitations. First, the EONET dataset has a limited sample size and category richness. Compared with **wildfires**, other categories, such as **volcanoes** and **sea and lake ice,** contain far fewer records, so the analysis cannot be equally extended to all natural hazards.

Second, the FIRMS data used in this study are mostly concentrated in **North America**, which means the results may not have strong global generalizability. In addition, FIRMS records **fire detections** rather than independent wildfire events. A single large wildfire may produce many fire points, and some detected points may represent small-scale burning rather than major wildfire disasters.

Finally, this project matches FIRMS and EONET wildfire records mainly through approximate **latitude–longitude ranges**. Although this helps identify likely related events, the method is not fully precise and may lead to missing or repeated records. Future work could improve the study by using more balanced datasets and more accurate spatial matching methods.

V. Conclusion

This project finds that wildfire activity follows clear **spatial and seasonal** patterns, with events concentrated in specific regions globally and highly uneven in magnitude and intensity across U.S. states. By combining EONET event data with FIRMS satellite detections, the analysis shows **strong consistency between the two sources**, including a clear dominance of daytime detections and peak wildfire activity in late summer and early fall. Furthermore, **temperature** is strongly positively correlated with both wildfire frequency and intensity, indicating that seasonal climatic conditions are a key driver of wildfire behavior and helping explain when and where wildfires are most severe.